

EVALUATING DIGITAL HEALTH

INTERVENTIONS

(3rd course
Specialization)

week 1)

DATA

→ Data } important as they impact on health equity
digital health }

Targets intervention

Monitoring

Current challenges in widespread distribution of equitable Data :
Digital health

- lack of primary data
- " " coordination between systems
- delays in making data available
- data doesn't meet end-users' needs

DATA must be:

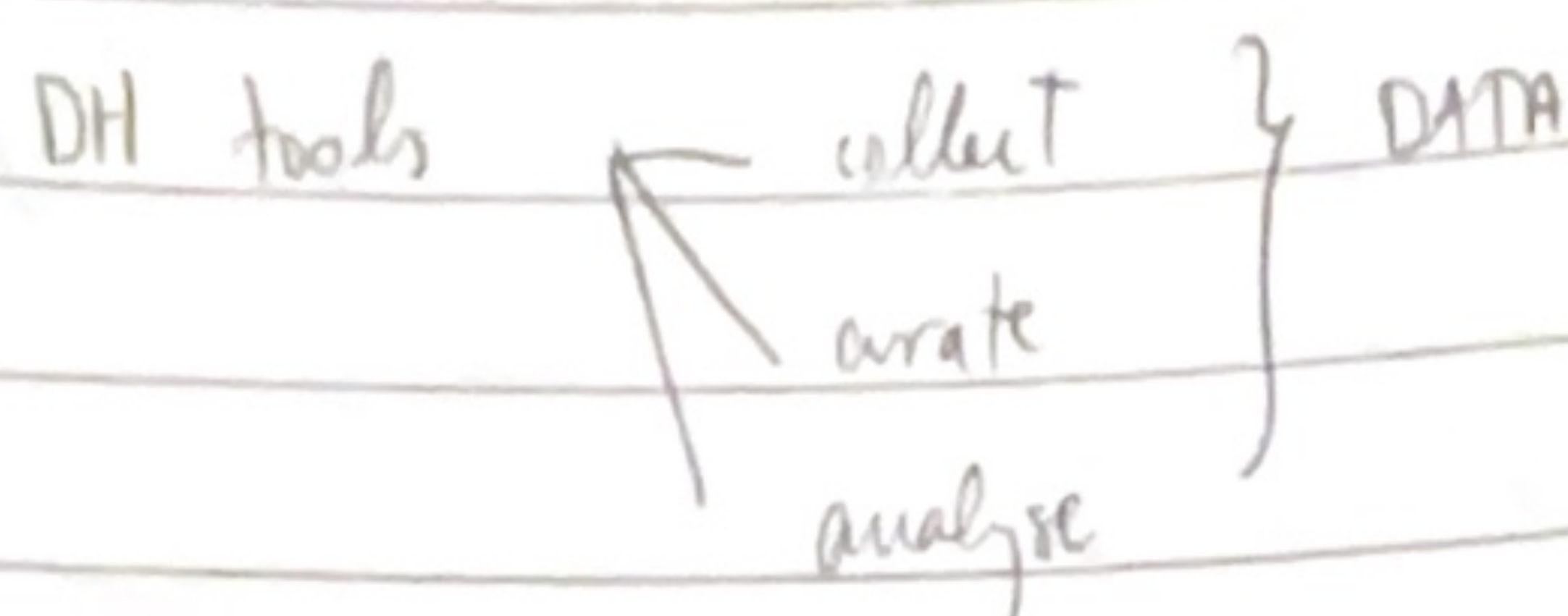
Possible actions:

- assessment of data gaps
- standardisation of data
- " " " collection tools
- alignment + coordination across systems
- tackle information flow problems
- investment in Digital health

timely
policy-relevant
reliable

maintain privacy
stored securely

→ Data is the currency of Digital health



→ Paper

≠ data need ≠ Statistical tests

→ Have the authors set the scene correctly?

- Are groups comparable?
- If necessary → are they adjusted for baseline differences?

Should be in:

age

sex distribution

key prognostic variables (ex size of tumour)

All Statistical tests are:

→ Parametric - they assume data were sampled from a particular form of distribution (eg normal distribution)

→ non-parametric

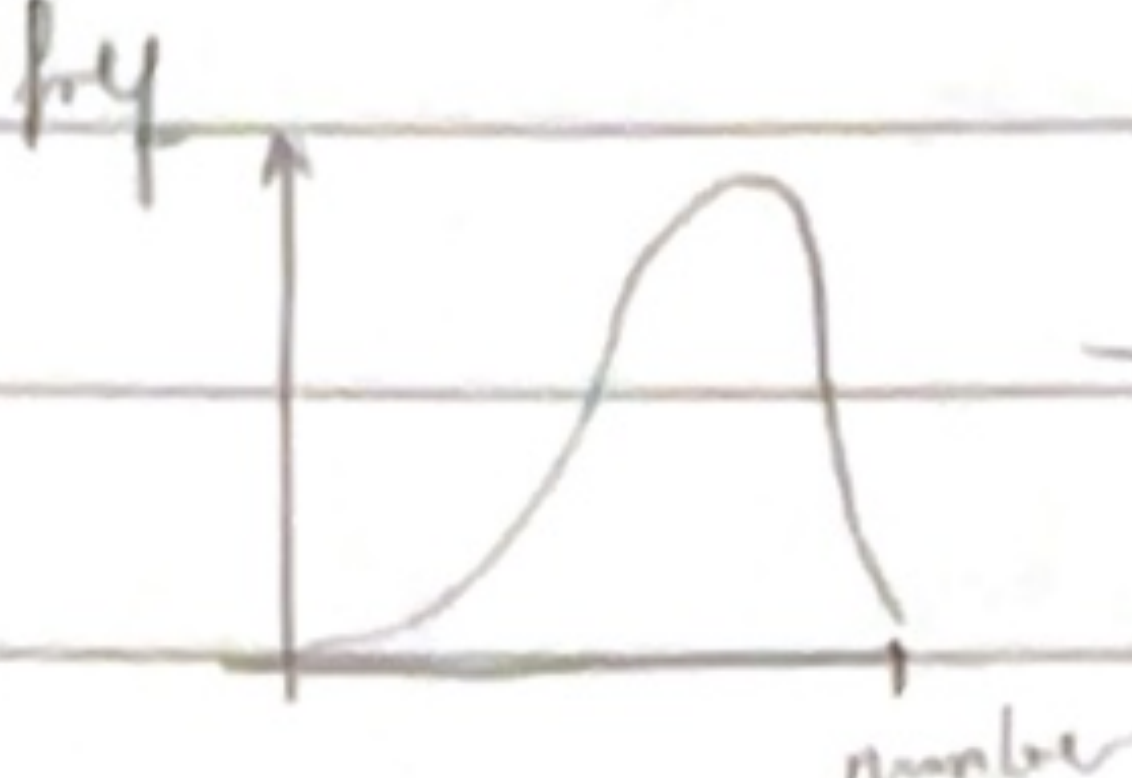


→ parametric tests are preferred (if possible) as they are more powerful

→ non-parametric analyse the rank order of values but not the absolute

↓
harder to use statistical test

→ Does the chosen test reflect the type of data analysed?

Skew distribution →  → this can be guessed through a math $f(x)$ to give a normal distribution (not all)

- Which Statistical test did the authors use? Are they appropriate?
 - If it's an unusual test → authors should give explanation + reference —
- the problem is when we use tests on the normal distribution to analyse non-normally distributed data → cheating

→ were the data analysed according to the original protocol?

→ were paired tests performed on paired data?
 — paired: 2 sets of values of the same thing / group / etc

→ was a two tailed test performed whenever the effect on a intervention could conceivably be a negative one?

tail — extremities of distribution
 the statistical analysis should test the hypothesis that either high / low values of the dataset were by chance
 ↓
 "two-tailed test"

→ Were outliers analysed with both common sense and appropriate statistical adjustments?

unexpected results might come from — subject itself

Statistically correcting for outliers → requires sophisticated analysis

errors ← measurement interpretation calculation

→ Only the one about the subject should be included in the analysis

Data Requirements and Considerations for Digital Health

both as input and output are fundamental

Is data being collected $\swarrow$ reliably? $\rightarrow$ Consider how data gathered
accurately? can be
VALIDATED

Validation

internal

external

numerator

denominator

Description

active efforts by a reporting facility to assess completeness and accuracy of data
survey + audit process by external agency
by performed by ext audit
" " " int "



then extracted



then transformed

$\rightarrow$ how will data be created into a form which enables valid analyses?

long term $\rightarrow$ STORAGE

$\swarrow$ fewer
sustainable

VISUALISE

Data MANAGEMENT

Acquiring / Validating / Storing / Protecting / Processing data

↓ to ensure

Accuracy / Reliability / timeliness

highly quality dataset

complete

otherwise

unique

accurate

consistent

↓
Data
Cleaning

before

Analysis

Data cleaning

→ no standardised way but recommendations:

- drop unwanted variables

- rename variables: short, explanatory names

Data analysing

→ start with calculating descriptive statistics from key variables -

continuous variables → mean / median /
standard deviation /
inter-quartile range

- histograms (assess skewness) (asymmetry)

Binary or categorical variables

tabulated

calculated

frequency
proportion

Establish associations → strength of association between outcome (dependent) variables:

t test if dep → cont

indep → binary

other explanatory
(independent) variables

derive p value - measure strength of the evidence
to reject null hypothesis

within
being the same as other -> chi squared
independent

Statistical model

depends on outcome variable

continuous	-> linear	calculate mean
binary	-> logistic	odd ratios
count	-> Poisson regression	rate ratios

Data Science Institute -> imperial (london)

big data -> authors created PolyglotHIS - polyglot - persistent
Health Information System
application

big data
look at
how components
work together

types of data
structured + unstructured - relational (PostgreSQL)
unstructured (e.g. lab images) - 2 NoSQL - MongoDB
containing relationships - patient - doctor
pt signs + DA
Neo4J

uses 3 Big Data
Storage types

User is exposed to 1 interface but has access to many databases

can be enhanced -> applying analytics to it's graphs
such as

segmentation

predictive modelling

-> system could identify individuals who might benefit from
preventive care and lifestyle changes

Pro-MED → Program for Monitoring emerging disease

collator user submitted reports of infectious diseases and toxins

Data is text → methods: text mining

Records are pre-processed using Natural Language toolkit (software)

- key words are extracted → compared and gauged
- co-occurrences compiled into time periods

2014-2016 Ebola epidemic

2015 Zika "

Article

keyword extraction methods ← machine learning base - supervised statistics based learning methods

simple statistics (TF-IDF)

topic based (Topic Modelling)

graph based (TextRank)

rely on labelled datasets

rare resource

+ In every Data Science / Machine Learning project

↓
Exploratory Data Analysis (EDA) → vital step to understand the data

↓
pick up relations that is hard to detect in a table

variables
both dep & indep are either discrete or categorical → chi squared

Statistical model

depends on outcome variable

continuous → linear

binary → logistic

count → Poisson regression

calculate mean $\neq$

odd ratios

rate ratios

Data Science Institute → imperial (london)

Big data →

actions created

PolyglotHIS

polyglot - persistent Health Information System application

big data

look at

how components work together

type of data

structured + financial

semistructured (e.g. lab images)

containing relationships patient-doctor

pt symp + DG

uses 3 Big Data

Storage types

- relational (PostgreSQL)

- 2 NoSQL → MongoDB

Neo4J

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Text data is subject to issues such as

- geographical errors
- ≠ spellings
- regional ≠ is → terminology notation
- temporal changes is → terminology notation

these
can be addressed by

pre-processing text data — standardise terms